Tyler J. Stennett

404-992-9681 | tyler.j.stennett@gmail.com | tylerstennett.com | github.com/tylerstennett

Research Summary

I am a Ph.D. student in Computer Science at the Georgia Institute of Technology working on AI for automated software test generation. My research investigates how the expressiveness and semantic understanding of LLMs can be combined with the rigor of program analysis and formal methods to produce higher quality, more comprehensive tests. My work has been published at ICSE, ISSTA, and ASE.

Education

Georgia Institute of Technology — Atlanta, GA
Doctor of Philosophy in Computer Science — *Jan 2026 – Present*
GPA: 4.00/4.00 — Research Focus: AI for Software Testing — Advisor: Alessandro Orso

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Computer Science — *Aug 2022 – May 2025*
GPA: 3.98/4.00 — Awards: Faculty Honors, Dean's List — Thesis Advisor: Thomas Plötz

Undergraduate Thesis: Layout-Agnostic Change Point Detection for Enhanced Human Activity Recognition in Smart Homes

Research Experience

Graduate Research Assistant — Aug 2025 – Present
Georgia Tech Software Engineering Lab — *Atlanta, GA*
- Lead research on novel abstractions and representations of test cases that improve how LLM agents comprehend, customize, and generate tests
- Designed and evaluated multi-agent test generation frameworks that ground agents in static analysis and decompose testing into iterative, supervised localization and composition agents
- Proposed techniques combining formal methods, graph representations, and LLMs for comprehensive test generation from abstract natural language

Research Scientist Intern — May 2026 – Aug 2026
IBM Research — *Yorktown Heights, NY*
- Conducted the first large-scale characterization of developer-written web API tests by extending the Hamster methodology, laying the groundwork for future research on API testing (submitted to ICSE 2027)
- Launched three ongoing research projects: natural-language-to-test generation, an empirical study of agents operating in test environments, and an investigation of fault pinning in existing testing techniques

Research Scientist Intern — May 2025 – Aug 2025
IBM Research — *Yorktown Heights, NY*
- Co-led the first large-scale, static analysis-driven characterization of Java test methods using JavaParser and Tree-sitter, informing future directions in automated software testing (**Hamster**, ICSE-SEIP 2026)
- Designed a multi-agent workflow that combines static analysis and LLMs to construct complex test cases from natural-language test descriptions (**Sakura**, ISSTA 2026)

Undergraduate Research Assistant — Aug 2023 – Aug 2024
Georgia Tech Software Engineering Lab — *Atlanta, GA*
- Developed **AutoRestTest**, a REST API testing framework combining multi-agent reinforcement learning (MARL), LLMs, and graph-based dependency modeling, improving code coverage by over 14% versus state-of-the-art tools
- Implemented **RESTGPT**, which uses LLMs to enrich OpenAPI specifications with operation constraints and example values for automated testing tools, raising rule identification from 79% to 97%

Honors & Awards

Future Leader, Doctoral Consortium, ACM AI Leadership Summit, 2026

Publications

(*) denotes equal contribution.

Rangeet Pan, **Tyler Stennett**, Divya Sankar, Bridget McGinn, Alessandro Orso, Raju Pavuluri, Saurabh Sinha, Maja Vukovic
Tangent: An Empirical Study of Testing Practices for LLM-Based Agent Applications
To appear in Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE), 2026.

Tyler Stennett, Rangeet Pan, Bridget McGinn, Alessandro Orso, Saurabh Sinha
Sakura: An Approach for Generating Complex Tests from Natural Language Test Descriptions
To appear in Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2026.

Tyler Stennett, Myeongsoo Kim, Saurabh Sinha, Alessandro Orso
AutoRestTest at the SBFT 2026 Tool Competition
In Proceedings of the 19th IEEE/ACM International Workshop on Search-Based and Fuzz Testing (SBFT), co-located with ICSE, 2026.

Rangeet Pan*, **Tyler Stennett***, Raju Pavuluri, Nate Levin, Alessandro Orso, Saurabh Sinha
Hamster: A Large-Scale Study and Characterization of Developer-Written Tests
In Proceedings of the 48th International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP), 2026.

Rangeet Pan, Raju Pavuluri, Ruikai Huang, Rahul Krishna, **Tyler Stennett**, Alessandro Orso, Saurabh Sinha
SAINT: Service-level Integration Test Generation with Program Analysis and LLM-based Agents
In Proceedings of the 48th International Conference on Software Engineering (ICSE), 2026.

Tyler Stennett, Myeongsoo Kim, Saurabh Sinha, Alessandro Orso
AutoRestTest: A Tool for Automated REST API Testing Using LLMs and MARL
In Proceedings of the 47th International Conference on Software Engineering: Demonstrations Track (ICSE-Demo), 2025.

Myeongsoo Kim, **Tyler Stennett**, Saurabh Sinha, Alessandro Orso
A Multi-Agent Approach for REST API Testing with Semantic Graphs and LLM-Driven Inputs
In Proceedings of the 47th International Conference on Software Engineering (ICSE), 2025.

Myeongsoo Kim, **Tyler Stennett**, Dhruv Shah, Saurabh Sinha, Alessandro Orso
Leveraging Large Language Models to Improve REST API Testing
In Proceedings of the 46th International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER), 2024.

Teaching Experience

Graduate Teaching Assistant, CS 6300: Software Development Process Aug 2024 – Present
Georgia Institute of Technology *Atlanta, GA*

- Led weekly office hours and gave guest lectures on automated test generation and software testing research
- Designed and maintained the autograder, including a static analysis tool that detects input-argument dumping

Professional Service

Journal Reviewer: ACM Transactions on Software Engineering and Methodology (TOSEM), Empirical Software Engineering (EMSE)

Technical Skills

Languages: Python, Java, C/C++, SQL, C#, JavaScript, HTML/CSS
Relevant Coursework: Programming Languages, SAT/SMT Solvers, Artificial Intelligence, Machine Learning, Natural Language Processing